

The Bucket Feature-Derivation Catalog

Every field the platform holds on one volume-bucket candlestick, everything derivable from it alone and in combination, the full Mode-10 selection feature set under three scopes, and the outcome/excursion side -- **every parameter coded**: this is the frozen registry the TP-vs-SL study implements from.

Source of truth: **smc_quant_legacy** @ pipeline-integrity | quant_engine - bucket_state - persistence - stats_overlay - heatmap - terminal (Mode 10) - config | **72 entries - 724 derivations - 22 groups (112) - 11 panels x 3 scopes + stats (387) - context/outcome/baselines (49) = 1281 coded fields** | Generated 2026-07-02

P	Persisted raw field (history.db)	D	Derived -- pure function, offline bit-identical	W	Windowed -- trailing 10/30/240 context
CTX	Context -- engine / structure state	6h	6-hour store -- depth.db / tape only	S	Selection-scoped display measure

How to read it: Part I lists each entry with a numbered set of derivations; Part II derives from GROUPS of entries (2+). Part III is the Mode-10 selection feature set -- the same panel + stats fields extracted under THREE scopes (B- base, J- journey, X- excursion). Part IV carries outcome/excursion labels and the offline KC/POC baselines. **Every field has a frozen code** (E05.03, G11.2, B-P4.02, S.05, C.01, O.06, K.05 ...) -- the study's CSV columns and results reference these codes verbatim. Items marked (6h) need the depth/tape stores; items marked (study join) belong to evaluation, not entry features.

A. Identity & Geometry

The bucket's footprint in time and price. Buckets close on a VOLUME target, never a clock -- so even the timestamps carry pace information.

E01 start_time	P	Epoch timestamp of the first trade that seeded the bucket.
E01.01 UTC hour-of-day (0-23)		E01.07 Top-of-hour or half-hour flag (algo-clock effects)
E01.02 Trading session tag: Asia / London / NY / overlap		E01.08 Minutes since UTC midnight (day position)
E01.03 Session-OPEN proximity flag: within 30m of London or NY open (Y/N)		E01.09 US data-hour flag (12:30 / 14:00 UTC class times)
E01.04 Session-CLOSE proximity flag (Y/N)		E01.10 Minutes since prior session's close
E01.05 Binance funding-window proximity: 00/08/16 UTC (Y/N + minutes)		E01.11 Same-hour historical norm join key
E01.06 Day-of-week / weekend flag		E01.12 Month / quarter position (calendar seasonality key)

E02 end_time	P	Epoch timestamp at close, i.e. the tick that filled the volume target.
E02.01 UTC hour-of-day at close		E02.06 Day-of-week at close / weekend flag
E02.02 Session tag at close		E02.07 Gap to NEXT bucket's start (tape idle time)
E02.03 Session-STRADDLE flag: opened in one session, closed in another (Y/N)		E02.08 Elapsed fraction of the current session
E02.04 Funding-timestamp straddle flag (crossed 00/08/16 UTC) (Y/N)		E02.09 Proximity to daily (00:00 UTC) candle close
E02.05 Round-hour close proximity (Y/N)		E02.10 Close inside a scheduled-news window flag

E03 duration	D	end_time - start_time. Volume buckets stretch in quiet tape and compress in fast tape.
E03.01 Raw seconds / log-duration		E03.06 Duration vs same-UTC-hour historical norm
E03.02 Duration percentile vs trailing 30 and 240 buckets		E03.07 Fast/slow STREAK length (consecutive buckets same class)
E03.03 Duration z-score vs trailing window		E03.08 Duration acceleration: ratio vs prior bucket
E03.04 FAST-fill flag: bottom decile (Y/N)		E03.09 Implied tape-pace regime tag (quiet/normal/hot)
E03.05 SLOW-fill flag: top decile (Y/N)		E03.10 Duration share of the trailing-10 total (burstiness)

E04 open / high / low / close	P	Classic OHLC of the bucket; high/low fall back to close on a pre-tick live snapshot.
E04.01 Gap vs prior close: size, sign, gap-in-ticks		E04.07 Equal-highs / equal-lows vs recent extremes (resting-liquidity pools)
E04.02 Round-number proximity of each price (xx00 / xx50)		E04.08 True range (gap-adjusted) vs raw range
E04.03 New N-bucket HIGH flag / new N-bucket LOW flag		E04.09 Log return open->close
E04.04 Relation to prior bucket: inside / engulfing / breakout tag		E04.10 Close vs POC side (above/below the control price)
E04.05 Close>open sign (up/down bucket)		E04.11 Distance of close to nearest round number in ticks
E04.06 Close position within range, 0..1		
E05 range	D	high - low. Denominator for sweep, fail, reclaim and moved fractions.
E05.01 Absolute range / range in ticks		E05.06 EXPANSION flag: range > 1.20 x ATR (Y/N)
E05.02 Range percentile vs trailing 30/240		E05.07 Expanding/contracting range STREAK
E05.03 Range z-score		E05.08 Range per second (volatility pace)
E05.04 Range vs ATR ratio (the compression/expansion read)		E05.09 Share of the day's range contributed so far
E05.05 COMPRESSION flag: range < 0.60 x ATR (Y/N)		E05.10 Range rank within the session
E06 result (body)	D	close - open: what all that effort actually moved.
E06.01 Sign (direction) and magnitude		E06.06 Body ratio vs prior body (momentum of result)
E06.02 Log return of the body		E06.07 Net progress over trailing N buckets (drift context)
E06.03 Body percentile / z-score vs window		E06.08 Marubozu-ness: body ~ range (Y/N)
E06.04 Same-direction streak length		E06.09 Body-in-ticks vs fee+spread hurdle (economically meaningful move?)
E06.05 Reversal flag vs prior bucket direction (Y/N)		E06.10 Contribution sign to trailing bucket-CVD of price
E07 body fraction	D	abs(close-open)/range -- the MOVED ramp (0.55-0.80): decisive bar vs round trip.
E07.01 Raw fraction 0..1		E07.06 Two-sided rejection flag (small body, both wicks large)
E07.02 TREND-BAR flag: fraction > 0.80 (Y/N)		E07.07 Conviction class: low / mid / high
E07.03 DOJI flag: fraction < 0.10 (Y/N)		E07.08 Fraction x volume interaction (decisive AND heavy?)
E07.04 Percentile / z vs trailing window		E07.09 Rolling-mean fraction (regime trendiness)
E07.05 Trend-bar streak length		E07.10 State-engine MOVED ramp value (0.55-0.80)
E08 upper / lower wick	D	Rejection tails: high-max(o,c) and min(o,c)-low.
E08.01 Upper wick size / lower wick size (ticks)		E08.06 Wick-to-body ratio per side
E08.02 Upper/range and lower/range fractions		E08.07 Rejected side inference (which extreme got refused)
E08.03 Wick asymmetry: (upper-lower)/range		E08.08 Wick percentile vs trailing window
E08.04 PIN-BAR flag: one wick > 60% of range (Y/N)		E08.09 Wick at NEW extreme flag (swept then rejected)
E08.05 BOTH-WICKS flag: whipsaw shape (Y/N)		E08.10 Wick into round-number / OB zone flag
E09 poc_price	P	Price level holding the largest footprint volume (Point of Control), finalized at close.
E09.01 POC round-number proximity		E09.06 Distance from POC to nearest order block
E09.02 POC shift vs prior bucket's POC (migration size + direction)		E09.07 Naked/untested prior POCs above and below (count, nearest)
E09.03 POC gap vs current close (ticks)		E09.08 POC retest count over trailing window
E09.04 POC at bucket extreme flag (top/bottom 15% of range)		E09.09 POC percentile within the day's range
E09.05 POC inside prior bucket's range flag (acceptance)		E09.10 POC coincides with a heatmap wall price (6h) (Y/N)
E10 poc_pos	D	POC position inside the range 0..1; stacked-high POC (0.50-0.85 ramp) is a bull-trap factor.
E10.01 Raw 0..1 position		E10.06 Trap-geometry ramp value (0.50-0.85)
E10.02 HIGH-POC flag (>0.70) / LOW-POC flag (<0.30)		E10.07 Persistently-high/low POC streak
E10.03 CENTER-POC flag (0.40-0.60, balanced auction)		E10.08 POC-pos flip vs prior bucket (rotation of control)
E10.04 Percentile vs trailing window		E10.09 Profile shape inference: p / b / D-shaped
E10.05 POC-pos vs close-pos divergence (control vs settle)		E10.10 POC-pos x direction interaction (control against the move?)

B. Volume & Aggression Flow

Who initiated. Taker side is exact at ingest, not inferred.

E11 curr_vol	P	Total bucket volume -- the defining quantum, ~= target_vol at close.
E11.01 Overshoot ratio: curr_vol / target_vol (last-print spillover)		E11.06 Volume per tick of range (grind density)
E11.02 Volume percentile / z vs trailing window		E11.07 Same-UTC-hour normalized volume (session-relative heat)
E11.03 Log volume		E11.08 Cumulative session volume position
E11.04 Share of trailing-N total volume (activity concentration)		E11.09 Day-rank of the bucket's volume
E11.05 Above/below-mean volume streak		E11.10 Volume regime tag (quiet / normal / heavy)

E12 buy_vol	P	Volume where the aggressor was the buyer (taker buys).
E12.01 Buy share: buy_vol / curr_vol		E12.06 Day-rank of buy volume
E12.02 Buy z-score / percentile vs trailing window		E12.07 Extreme-buy flag: top decile (Y/N)
E12.03 Buy-dominance streak (share > 0.5)		E12.08 Log buy volume
E12.04 Change in buy share vs prior bucket		E12.09 Buy share vs same-hour norm
E12.05 Buy volume per second (buy pace)		E12.10 Buy-share slope over trailing 5 buckets

E13 sell_vol	P	Volume where the aggressor was the seller (taker sells).
E13.01 Sell share: sell_vol / curr_vol		E13.06 Day-rank of sell volume
E13.02 Sell z-score / percentile		E13.07 Extreme-sell flag: top decile (Y/N)
E13.03 Sell-dominance streak		E13.08 Log sell volume
E13.04 Change in sell share vs prior bucket		E13.09 Sell share vs same-hour norm
E13.05 Sell volume per second		E13.10 Sell-share slope over trailing 5 buckets

E14 delta	D	buy_vol - sell_vol: net aggression in contracts.
E14.01 Sign and magnitude		E14.06 Delta acceleration (this vs prior magnitude)
E14.02 Delta as % of volume		E14.07 Trailing-N cumulative delta (bucket CVD)
E14.03 Delta z-score / percentile		E14.08 Delta vs price divergence flag (delta up, close down / inverse)
E14.04 Same-sign delta streak (pressure persistence)		E14.09 Day-rank of abs(delta)
E14.05 Delta FLIP flag vs prior bucket (Y/N)		E14.10 Intrabucket delta path min/max (6h tape)

E15 delta fraction	D	abs(delta)/curr_vol -- the DELTA ramp (0.10-0.55): how one-sided initiative was.
E15.01 Raw fraction		E15.06 Alignment with body fraction (initiative translated?)
E15.02 ONE-SIDED flag: > 0.55 (Y/N)		E15.07 Fraction regime mean (trailing)
E15.03 BALANCED flag: < 0.10 (Y/N)		E15.08 Session rank of one-sidedness
E15.04 Percentile / z vs window		E15.09 Fraction x volume interaction (one-sided AND heavy)
E15.05 One-sidedness streak		E15.10 DELTA ramp value fed to the state engine

C. Open-Interest Attribution (Step 19)

The bucket's volume decomposed by what it did to open interest -- initiative vs unwind.

E16 opL	P	Volume attributed to FRESH LONGS opening.
E16.01 opL / vol fraction		E16.06 Day-rank of opL
E16.02 z-score / percentile vs window		E16.07 opL into highs vs into lows (location context)
E16.03 FRESH-LONG-WAVE flag: fraction > 0.30 (Y/N)		E16.08 opL acceleration vs prior bucket
E16.04 opL streak (persistent long initiative)		E16.09 Share of total opens (opL/(opL+opS))
E16.05 opL vs opS dominance ratio		E16.10 TRAP-FUEL candidate: heavy opL into a failed upside break

E17 opS	P	Volume attributed to FRESH SHORTS opening.
E17.01 opS / vol fraction		E17.06 Day-rank of opS
E17.02 z-score / percentile		E17.07 opS into lows vs into highs (location context)
E17.03 FRESH-SHORT-WAVE flag: fraction > 0.30 (Y/N)		E17.08 opS acceleration
E17.04 opS streak		E17.09 Share of total opens
E17.05 opS vs opL dominance ratio		E17.10 TRAP-FUEL candidate: heavy opS into a failed downside break
E18 cLL	P	Volume attributed to LONGS CLOSING (unwind into the tape).
E18.01 cL / vol fraction		E18.06 Share of total closes (cL/(cL+cS))
E18.02 z / percentile		E18.07 cL acceleration
E18.03 LONG-CAPITULATION flag: heavy cL into lows (Y/N)		E18.08 Day-rank of cL
E18.04 cL streak (sustained unwind)		E18.09 cL at NEW low flag (puke print)
E18.05 Long-cohort net: opL - cL (cohort building or leaving)		E18.10 Closing-flow ramp contribution (CLOSE_FLOW 0.15-0.50)
E19 cLS	P	Volume attributed to SHORTS CLOSING (short covering).
E19.01 cS / vol fraction		E19.06 Share of total closes
E19.02 z / percentile		E19.07 cS acceleration
E19.03 SHORT-COVER-POP flag: heavy cS into highs (Y/N)		E19.08 Day-rank of cS
E19.04 cS streak		E19.09 cS at NEW high flag (forced-out print)
E19.05 Short-cohort net: opS - cS		E19.10 Closing-flow ramp contribution
E20 churn	P	OI-neutral transfer volume: hands change, net interest does not.
E20.01 churn / vol fraction		E20.06 Churn x moved interaction: transfer that MOVED price (markdown/markup hand-off)
E20.02 ROTATION flag: fraction > 0.55 (CHURN ramp) (Y/N)		E20.07 Day-rank of churn
E20.03 z / percentile		E20.08 Churn acceleration
E20.04 High-churn streak		E20.09 OI-neutrality corroboration (dOI small while churn high)
E20.05 Churn regime mean (trailing)		E20.10 Hand-off inference tag: distribution vs accumulation candidate
E21 delta_oi	D	Net open-interest change: opens minus closes (stats-box dOI).
E21.01 Sign: BUILD vs DRAIN		E21.06 OI-NEUTRAL flag: dOI /vol < 0.22 (Y/N)
E21.02 Magnitude / vol fraction		E21.07 BUILD ramp value (0.08-0.18)
E21.03 z / percentile		E21.08 dOI acceleration
E21.04 BUILD streak / DRAIN streak lengths		E21.09 Trailing-N cumulative dOI (positioning trend)
E21.05 dOI x delta quadrant: new-money push / covering push / distribution / unwind-drop		E21.10 Day-rank of dOI
E22 open/close/churn fractions	D	The named state-engine ramps computed from the decomposition.
E22.01 OPEN_OI ramp value (0.10-0.50)		E22.06 BUILD ramp value (0.08-0.18)
E22.02 DRAIN_OI ramp value (0.05-0.45)		E22.07 Dominant-component label (which fraction leads)
E22.03 CLOSE_FLOW ramp value (0.15-0.50)		E22.08 Component ENTROPY (how mixed the bucket's OI story is)
E22.04 CHURN ramp value (0.55-0.90)		E22.09 Open:close ratio
E22.05 OI_NEUTRAL ramp value (0.05-0.22)		E22.10 Long-side vs short-side initiative index

D. Effort / Result (Step 4)

How much aggression it took to move price; one scale for live and closed buckets.

E23 buyer_er	P	Buyer effort: buy_vol per unit of volume-weighted price dispersion.
E23.01 Raw value / log value		E23.06 Hot streak length
E23.02 z vs trailing-30 (the baseline convention)		E23.07 buyer_er / seller_er ratio (effort dominance)
E23.03 Multiple vs trailing mean (stats-box x-figure)		E23.08 Effort acceleration vs prior
E23.04 Percentile vs window		E23.09 Day-rank of buyer effort
E23.05 HOT flag: >= 1.5x baseline (Y/N)		E23.10 Effort-hot leg value for exhaustion gating

E24 seller_er	P	Seller effort: sell_vol per unit of dispersion -- the symmetric read.
E24.01 Raw value / log value		E24.06 Hot streak
E24.02 z vs trailing-30		E24.07 seller_er / buyer_er ratio
E24.03 Multiple vs trailing mean		E24.08 Effort acceleration
E24.04 Percentile		E24.09 Day-rank
E24.05 HOT flag: >= 1.5x (Y/N)		E24.10 Effort-hot leg value for exhaustion gating

E25 effort_ticks	D	The dispersion denominator: volume-weighted footprint spread (1-tick floor).
E25.01 Raw dispersion (ticks)		E25.06 Narrow/wide streak
E25.02 Percentile / z vs window		E25.07 Dispersion per unit volume
E25.03 NARROW-auction flag (bottom quintile)		E25.08 Regime mean of dispersion
E25.04 WIDE-auction flag (top quintile)		E25.09 Auction-type inference: compressed grind vs sweep
E25.05 Dispersion vs range ratio (profile fatness)		E25.10 Floor-hit flag (dispersion at the 1-tick floor)

E26 B-E/R, S-E/R anomaly	W	Each side's E/R as a multiple of its trailing-30 baseline.
E26.01 Buyer multiple / seller multiple values		E26.06 First-hot-after-quiet flag (regime break)
E26.02 Hotter side label		E26.07 Multiple slope over last 5 buckets
E26.03 BOTH-HOT flag (Y/N)		E26.08 Side-switch of hotness flag
E26.04 Multiple percentile vs day		E26.09 Effort-hot x no-result divergence candidate (pair with body)
E26.05 Hot-side streak		E26.10 Anomaly co-fire with velocity flag (Y/N)

E27 E/R neon border flag	W	Descriptive 2px border when a side's E/R z is strongly elevated.
E27.01 Fired (Y/N) + side		E27.06 With-trend vs counter-trend fire
E27.02 z magnitude behind the fire		E27.07 Consecutive-fire streak
E27.03 Count of fires in trailing window (density)		E27.08 Co-fire with abnormal-velocity flag
E27.04 First-fire-in-N flag		E27.09 Co-fire with liquidation presence
E27.05 Fired AT a new extreme flag		E27.10 Fire followed by direction flip (study join)

E. Pace & Velocity

How fast the bucket filled, absolutely and versus its own recent norm.

E28 vol_mult	P	Close-time velocity as a multiple of the engine's rolling average (Speed x).
E28.01 Raw multiple		E28.06 Speed streak (consecutive > 1x)
E28.02 STRONG flag: >= 1.2 (stats-box highlight)		E28.07 Speed acceleration vs prior
E28.03 Squeeze-violence ramp value (1.30-3.00)		E28.08 QUIET flag: < 1.0 (Y/N)
E28.04 Mild-pace ramp value (1.00-2.20)		E28.09 Day-rank of speed
E28.05 Percentile / z vs window		E28.10 Speed x direction interaction (fast up vs fast down)

E29 vel	D	curr_vol / duration in units per second -- the raw pace number.
E29.01 Raw units/s / log pace		E29.06 Day-max pace flag (Y/N)
E29.02 Percentile / z vs trailing window		E29.07 Pace regime class (quiet/normal/hot)
E29.03 Same-UTC-hour normalized pace		E29.08 Pace vs range coupling (fast AND wide?)
E29.04 Pace streak		E29.09 Pace vs spread regime (6h BBO join)
E29.05 Pace acceleration		E29.10 Trailing-30 mean pace (the flag's denominator)

E30 abnormal-velocity flag	W	Fires at vel $\geq 5.0x$ trailing-30 mean (caps at $10x$); ~6% of buckets. Visual-study marker, not a signal.
E30.01 Fired (Y/N)		E30.06 Fire direction (bucket up/down)
E30.02 Ratio value ($5x..10x+$)		E30.07 Fired AT a sweep flag
E30.03 CAPPED flag ($\geq 10x$)		E30.08 Co-fire with liquidation flag
E30.04 Fires count in trailing window (cluster density)		E30.09 Co-fire with E/R neon flag
E30.05 First-after-quiet flag		E30.10 Cluster id / time since previous fire

E31 avg_velocity	CTX	The engine's rolling velocity baseline all multiples are measured against.
E31.01 Current level		E31.06 Time since last regime change
E31.02 Slope over trailing window (regime drift)		E31.07 Ratio to day mean
E31.03 Stability (variance) of the baseline		E31.08 Session-profile position (hour curve)
E31.04 Percentile vs the day		E31.09 Baseline freshness (window fill)
E31.05 Jump flag (recent step-change)		E31.10 Cross-check: implied buckets/hour

E32 target_vol	CTX	The adaptive volume quantum a bucket must accumulate to close.
E32.01 Current size		E32.06 Implied buckets/hour at current pace
E32.02 Recalibration delta (latest change)		E32.07 Percentile vs history
E32.03 Slope of target over the day		E32.08 Jump flag (regime resize)
E32.04 Ratio to day-mean target		E32.09 Timeframe-relative scale (1m vs 5m ladder)
E32.05 Stability (variance)		E32.10 Time since last recalibration

F. Exhaustion (Step 5)

Effort spiking while OI drains and closing flow powers the move: participants leaving, not arriving. Validated 36x separation from strong continuation.

E33 b_mult / s_mult	W	Per-side effort z-multipliers over the 30-bucket EXH_WINDOW.
E33.01 Buyer multiplier / seller multiplier values		E33.06 Slope over the selection window
E33.02 Dominant side label		E33.07 Both-elevated flag (Y/N)
E33.03 ABSORB-fire read: ≥ 1.20 ramp (1.20-2.00)		E33.08 Crossover proximity (bull vs bear lines)
E33.04 TRANSLATE read ramp (1.30-2.00)		E33.09 Stability of the z-baseline (window health)
E33.05 Percentile of each vs day		E33.10 Multiplier at bucket extremes vs mid

E34 gated true-exhaustion	D	Geomean of effort-hot AND OI-draining AND closing-flow; any missing leg gates to zero.
E34.01 Score 0..1 + side		E34.06 Fired streak / decay since last fire
E34.02 FIRED flag above threshold (Y/N)		E34.07 First-fire-in-N flag
E34.03 Weakest-leg name (what limited the score)		E34.08 Fired AT new extreme flag
E34.04 Per-leg values (3 features)		E34.09 Co-fire with reclaim (squeeze-completion candidate)
E34.05 Percentile of score vs window		E34.10 Score slope across the last 5 buckets

E35 exhaustion lines + crossover	S	Selection-scoped expanding-z lines, symmetric smoothing, crossover diamonds ('1' toggle).
E35.01 Crossover occurred in selection (Y/N) + direction		E35.06 Confirmed vs forming state of latest cross
E35.02 Diamond count in the selection		E35.07 Cross at a price extreme flag
E35.03 Bull-line slope / bear-line slope		E35.08 Double-cross whipsaw flag
E35.04 Spread between the two lines (dominance margin)		E35.09 Line values at the selected bucket
E35.05 Buckets since last cross		E35.10 Alignment with the state verdict (Y/N)

G. Absorption & Effective Aggression

The proven partition: the share of volume that failed to move price, and the share that did the moving.

E36 suppression score s	D	0..1: how much of the bucket's aggression price refused to translate.
E36.01 Raw s		E36.06 s x delta-fraction interaction (one-sided AND absorbed = wall)
E36.02 VALIDATED flag: s >= 0.60 (Y/N)		E36.07 Absorbed-side inference (with delta sign)
E36.03 Percentile / z vs window		E36.08 s regime mean
E36.04 High-s streak		E36.09 s at price extreme flag (absorption at highs/lows)
E36.05 s slope over last 5 buckets		E36.10 Rank of s within the day

E37 bull/bear absorption (V*s)	D	Volume absorbed, split by the side under attack.
E37.01 Absolute absorbed volume + side		E37.06 Absorption acceleration
E37.02 Percentile / z vs window		E37.07 Adaptive-zone crossing flag (display zones)
E37.03 Day-rank of absorbed volume		E37.08 Validated-strength dot present (Y/N)
E37.04 Absorbed share of trailing-N volume		E37.09 Absorption at extreme location flag
E37.05 Absorption streak per side		E37.10 Bull-vs-bear absorption dominance ratio

E38 effective aggression $V^*(1-s)$	D	The exact mirror: volume that actually moved price (partition proven).
E38.01 Absolute eff-agg + side		E38.06 Eff-agg streak per side
E38.02 Percentile / z		E38.07 Acceleration
E38.03 Day-rank		E38.08 Zone-crossing flag
E38.04 Eff-agg share of trailing-N volume		E38.09 Partition sanity: $V*s + V^*(1-s) == V$ (audit feature)
E38.05 Dominance ratio vs absorption		E38.10 Eff-agg per tick moved (thrust efficiency)

E39 validated-strength dot	D	The grounded s = 0.60 boundary: absorption strong enough to count.
E39.01 Present (Y/N) + side		E39.06 Co-fire with exhaustion fire
E39.02 Count in trailing window		E39.07 Co-fire with liquidation presence
E39.03 First-in-N flag		E39.08 Dot preceded direction flip (study join)
E39.04 Consecutive-dot streak		E39.09 Dot density regime (dots per 100 buckets)
E39.05 Dot at new high/low flag		E39.10 Side-switch of dots flag

E40 absorption/eff-agg zones	S	Adaptive display zones with floorless sliders over the stored raw values.
E40.01 In-zone (Y/N) per zone		E40.06 Breach direction of last exit
E40.02 Zone width (current adaptive setting)		E40.07 Zone percentile setting (slider state)
E40.03 Distance to nearest zone edge		E40.08 Dwell streak
E40.04 Dwell time in zone (buckets)		E40.09 Zone vs validated-dot agreement
E40.05 Zone entries count in selection		E40.10 Zone regime (tight/wide adaptive state)

E41 absorption marks	6h	Intrabucket iceberg-absorption events detected from live tape bands.
E41.01 Marks count in bucket		E41.06 Marks clustering (bursts vs spread)
E41.02 Side split of marks		E41.07 Mark-to-close displacement (did price leave the mark?)
E41.03 Largest mark size		E41.08 Marks per minute
E41.04 Marks at bucket extremes flag		E41.09 Mark co-location with a heatmap wall
E41.05 Mark price-band width		E41.10 Iceberg-side classification share (buy vs sell)

H. The Footprint Ladder (levels)

The complete per-price distribution -- anything a footprint can yield is computable from this field.

E42 levels {price:{b,s}}

P

Full ladder: buy and sell volume at every traded price.

E42.01 Traded-level count

E42.02 Top-3 level concentration share

E42.03 Volume-weighted mean price vs close (settle skew)

E42.04 Profile skewness (volume tilt)

E42.05 Profile kurtosis (fat middle vs barbell)

E42.06 Max single-level side imbalance (b:s odds)

E42.07 Single-print level count (thin spots)

E42.08 Profile shape class: P / b / D

E42.09 Low-volume-node presence + nearest LVN distance

E42.10 High-volume-node distance to close

E42.11 Buy-vs-sell tilt by range third (top/mid/bottom loading)

E43 POC volume / share

D

Volume at the control price and its share of the bucket.

E43.01 POC share of curr_vol

E43.02 Percentile / z of the share

E43.03 DOMINANT-POC flag (share > threshold)

E43.04 POC vs 2nd-node ratio (peakedness)

E43.05 Dominant-POC streak

E43.06 THIN-POC flag (weak control)

E43.07 Buy:sell split AT the POC

E43.08 POC-share regime mean

E43.09 POC share x poc_pos interaction (strong control at extreme)

E43.10 Day-rank of POC share

E44 per-level imbalance flag

W

Level side-volume vs the trailing-30 BER/SER baseline; strong levels invert to neon.

E44.01 Flagged-level count + side split

E44.02 Max imbalance multiple on the ladder

E44.03 STACKED flag: consecutive flagged levels (Y/N + depth)

E44.04 Stack side (buyer wall vs seller wall)

E44.05 Stack at bucket extreme flag

E44.06 Flag density percentile vs window

E44.07 First-stack-in-N flag

E44.08 Stack vs close side (did price respect the stack?)

E44.09 Neon count per side

E44.10 Diagonal-imbalance read across adjacent levels

E45 ladder depth / dispersion

D

Traded-level count and volume-weighted spread (the effort_ticks geometry).

E45.01 Level count / dispersion in ticks

E45.02 Dispersion percentile vs window

E45.03 FAT profile flag / THIN profile flag

E45.04 Dispersion vs range ratio

E45.05 z-score of dispersion

E45.06 Fat/thin streak

E45.07 Dispersion regime mean

E45.08 Auction-type tag: grind vs sweep

E45.09 Single-print tail flag (unfinished business at an extreme)

E45.10 Dispersion per unit volume

E46 further derivables

D

Not pre-computed by the platform, but exactly derivable from levels.

E46.01 Value-area (70%) width + position vs close

E46.02 VA overlap vs prior bucket (acceptance/rejection)

E46.03 Top-vs-bottom loading index

E46.04 Unfinished-auction flag at high / at low

E46.05 LVN / HVN map (counts + nearest distances)

E46.06 Profile entropy (auction dispersion)

E46.07 Delta-by-level tilt (where the net aggression sat)

E46.08 Mid-range acceptance share

E46.09 Poor-high / poor-low flags

E46.10 Composite profile tag for pattern joins

I. Forced Liquidations

Forced flow inside the bucket from the forceOrder stream (pre-feed buckets carry structural zeros).

E47 liq_short

P

Volume of SHORTS liquidated (forced buys).

E47.01 Absolute + /vol fraction

E47.02 z / percentile vs window

E47.03 FIRED flag: fraction >= 0.03 (LIQ ramp floor)

E47.04 VIOLENT flag: fraction >= 0.20

E47.05 Liq-short streak

E47.06 Day-rank

E47.07 Co-fire with up-move (squeeze corroboration)

E47.08 First-after-quiet flag

E47.09 Share of window's total liq-short

E47.10 Acceleration vs prior bucket

E48 liq_long	P	Volume of LONGS liquidated (forced sells).
E48.01 Absolute + /vol fraction		E48.06 Day-rank
E48.02 z / percentile		E48.07 Co-fire with down-move (long-squeeze corroboration)
E48.03 FIRED flag (≥ 0.03)		E48.08 First-after-quiet flag
E48.04 VIOLENT flag (≥ 0.20)		E48.09 Share of window's total liq-long
E48.05 Liq-long streak		E48.10 Acceleration

E49 liq_fractions	D	liq_side/vol and liq_total/vol -- the LIQ and LIQ_COIL ramps.
E49.01 Total-liq fraction		E49.06 Cluster count in trailing window
E49.02 Side-dominance of forced flow		E49.07 Buckets since last liq event
E49.03 COIL-band flag: absorbed liqs inside compression (0.03-0.15)		E49.08 Liq vs delta alignment (forced flow with or against)
E49.04 BOTH-SIDES liq flag		E49.09 Liq at price extremes flag
E49.05 Percentile of total fraction		E49.10 Cascade shape: rising liq sequence across buckets

J. Trade-Size Classification (since 2026-06-30)

Per-side size histograms over log-spaced bins; store raw, threshold live -- any cutoff applies retroactively.

E50 sz_cb / sz_cs	P	Trade COUNT per size bin, buy / sell.
E50.01 Counts per bin (vector feature)		E50.06 WHALE-print count (bins $\geq$ p99)
E50.02 Large-count share (bins $\geq$ p95 anchor)		E50.07 ZERO-LARGE flag (no large prints at all)
E50.03 Small-count share (bins $\leq$ -p50)		E50.08 Large-count percentile vs window
E50.04 Count entropy across bins		E50.09 Large-count streak
E50.05 Highest occupied bin (largest print class)		E50.10 Buy-vs-sell count skew per size class

E51 sz_vb / sz_vs	P	Trade VOLUME per size bin, buy / sell.
E51.01 Volume per bin (vector feature)		E51.06 Max-bin volume + its bin index
E51.02 LARGE-volume share of bucket		E51.07 z / percentile of large share
E51.03 WHALE-volume share		E51.08 Large-share streak
E51.04 SMALL-volume share		E51.09 Dominance flip of large volume side
E51.05 Side split of large volume		E51.10 Volume entropy across bins

E52 LARGE/SMALL/WHALE slices	D	Counts and volumes above/below any threshold, re-sliced instantly.
E52.01 Large net side: large-buy minus large-sell		E52.06 Whale present (Y/N)
E52.02 Large delta as % of bucket delta		E52.07 Large share of total delta
E52.03 Small net side		E52.08 Small-crowd one-sidedness
E52.04 LARGE-vs-SMALL divergence flag (opposed)		E52.09 Threshold-sweep stability: does the read hold p90->p97?
E52.05 Large-participation trend over trailing N		E52.10 Day-rank of large participation

E53 size_thr	CTX	Daemon rolling 60-min percentile anchors (p95 large / p99 whale / ~p50 small).
E53.01 Current p50/p95/p99 values		E53.06 Time since last material shift
E53.02 Anchor slope (sizing regime drift)		E53.07 Window fill n (cold-start health)
E53.03 p99/p50 ratio (live tail width)		E53.08 Bucket max print vs anchors (class of the biggest print)
E53.04 Anchor jump flag		E53.09 Anchor stability (variance)
E53.05 Percentile of anchors vs the day		E53.10 Cross-check vs stored histograms (audit)

K. VPIN & Flow Toxicity

Volume-synchronized probability of informed trading, adapted to SOL's real range.

E54 vpin	CTX	Rolling 50-bucket VPIN estimate (persisted in engine_state).
E54.01 Current value		E54.06 TOXIC-entry flag (crossed up into toxic)
E54.02 Tier: calm / warn / toxic (adaptive percentiles)		E54.07 VPIN vs delta alignment (toxic AND one-sided)
E54.03 z / percentile vs trailing 240		E54.08 Regime mean
E54.04 Slope over last N buckets		E54.09 Buckets since last toxic episode
E54.05 Tier-dwell streak		E54.10 VPIN x speed co-elevation flag

E55 adaptive toxic/warn tiers	W	Thresholds as PERCENTILES of the trailing-240 VPIN distribution (self-calibrating).
E55.01 Current warn / toxic threshold values		E55.06 Tier at bucket close
E55.02 Distance of vpin to toxic line		E55.07 Threshold drift (window freshness)
E55.03 Tier transitions count in window		E55.08 Tier stability score
E55.04 Dwell time per tier		E55.09 Percentile-window fill health
E55.05 Crossing direction of last transition		E55.10 Tier co-fire with liq/velocity flags

L. Windowed Structure Context

The bucket measured against its neighborhood -- all recomputable offline from the bucket sequence.

E56 ATR / range compression	W	Trailing range baseline; range/ATR (0.60-1.20) reads inside-bar compression.
E56.01 ATR value		E56.06 ATR percentile vs day
E56.02 range/ATR ratio		E56.07 Squeeze duration: consecutive compressed buckets
E56.03 COMPRESSION flag (< 0.60)		E56.08 Post-compression break direction (study join)
E56.04 EXPANSION flag (> 1.20)		E56.09 ATR regime tag
E56.05 ATR slope (vol regime drift)		E56.10 ATR x duration coupling (slow AND tight = coil)

E57 sweep depth	W	How far the bucket swept beyond prior 10-bucket extremes, over range (0.03-0.50 ramp).
E57.01 Swept prior HIGH (Y/N) / swept prior LOW (Y/N)		E57.06 Sweep at session extreme flag
E57.02 Sweep depth / range value		E57.07 Sweeps count in trailing window
E57.03 Depth percentile vs window		E57.08 Distance to nearest untouched equal-highs/lows pool
E57.04 BOTH-SIDES sweep flag		E57.09 Sweep-then-reclaim same-bucket flag
E57.05 Sweep magnitude z		E57.10 Sweep direction vs close direction (grab-and-go?)

E58 fail / reclaim depth	W	Failed-break depth (trap factor) and close-back-past-swept-level depth (squeeze reclaim).
E58.01 Fail depth /range value (0.05-0.50 ramp)		E58.06 Fail percentile vs window
E58.02 Fail direction (upside/downside break failed)		E58.07 Reclaim strength z
E58.03 Reclaim depth /range (0.00-0.30 ramp)		E58.08 Consecutive-fail streak
E58.04 Reclaim in SAME bucket flag (speed of reclaim)		E58.09 Close position after reclaim (conviction)
E58.05 Trap-geometry composite (fail x POC-pos x trapped-opens)		E58.10 Squeeze-floor trigger proximity (liq >= 0.10)

E59 trailing-30 baselines	W	The shared denominators: E/R baselines, velocity mean, exhaustion z-window.
E59.01 Each baseline level (BER, SER, vel-mean, exh-z mean/sd)		E59.06 Cross-metric coherence (all hot vs mixed)
E59.02 Baseline slopes (regime drift per metric)		E59.07 Baseline jump flags
E59.03 Baseline variance (stability)		E59.08 Position of window extremes (how long ago the max)
E59.04 Window fill / freshness n		E59.09 Baseline percentile vs the day
E59.05 Regime tags per metric		E59.10 Normalization health audit (zeros / floors hit)

M. The 12-State Verdict (classify_bucket)

One interpretive verdict per bucket with honest confidence; pure function -- every historical bucket can be labeled offline identically.

E60 state + confidence D Highest-scoring of 12 states; the geomean score IS the confidence (starred >= 0.80).

E60.01 State label (categorical) + one-hot vector
E60.02 Confidence value
E60.03 STARRED flag (>= 0.80)
E60.04 Confidence percentile vs window
E60.05 Same-state streak length

E60.06 Transition pair tag: prior state -> this state
E60.07 Directional class mapping: bullish / bearish / neutral
E60.08 State entropy over trailing window (regime clarity)
E60.09 Confidence slope over last N
E60.10 Verdict at price extreme flag

E61 factor breakdown D The named geomean inputs per state -- score and debug can never drift.

E61.01 Weakest-factor name (what limited the verdict)
E61.02 Weakest-factor value (gating margin)
E61.03 Runner-up state + score gap (fragility)
E61.04 Floored soft-factor count
E61.05 Squeeze-floor engaged (Y/N)

E61.06 Full factor vector as features
E61.07 Core-vs-soft balance index
E61.08 Factor coherence (variance across inputs)
E61.09 Hard-gate near-miss flags (core just above 0)
E61.10 Factor deltas vs prior bucket

E62 the 12 states D STRONG BULL/BEAR; BULL/BEAR EXHAUSTION; BULL/BEAR TRAP; SHORT/LONG SQUEEZE; LIQUIDITY COIL; ROTATION; CHOP; NEUTRAL.

E62.01 Historical frequency of each state
E62.02 Mean same-state run length
E62.03 Transition-matrix row for current state
E62.04 Per-state confidence distribution
E62.05 Session-hour distribution per state

E62.06 State co-occurrence with flags (liq / velocity / dots)
E62.07 Rarity flag (bottom-decile frequency state)
E62.08 State cluster tag (trend / reversal / neutral families)
E62.09 Per-state next-bucket outcome stats (study join)
E62.10 Regime dependence (state mix drift over days)

N. Market-Structure Context (order_blocks)

Where the bucket sits relative to detected structure (Otsu-thresholded zones, per timeframe).

E63 OB zones per tf CTX The active order-block set (price bands + metadata) for each of the 5 timeframes.

E63.01 Active-OB count per tf
E63.02 Nearest OB distance (ticks / ATRs)
E63.03 Nearest OB side: support vs resistance
E63.04 Nearest OB age (buckets since formed)
E63.05 Nearest OB strength score

E63.06 Freshly-formed OB flag (this session)
E63.07 Mitigated fraction of nearest OB
E63.08 Nearest OB width
E63.09 Stacked-OB count within N ticks
E63.10 OB density above vs below price

E64 distance-to-OB / inside-OB D The bucket's position relative to the nearest zone.

E64.01 Signed distance in ticks
E64.02 Distance in ATR units
E64.03 INSIDE-OB flag (Y/N) + which tf
E64.04 Approach velocity (distance change per bucket)
E64.05 FIRST-TOUCH flag of this OB

E64.06 Touch count of this OB so far
E64.07 Rejection-at-OB flag (wick into, close out)
E64.08 Close inside vs wick-only penetration
E64.09 Buckets since last touch
E64.10 BREAK-THROUGH flag (closed beyond the zone)

E65 cross-tf confluence CTX Order-block overlap scoring across timeframes -- the platform's original idea.

E65.01 Confluence score at current price
E65.02 Count of tfs overlapping here
E65.03 Highest timeframe involved
E65.04 Distance to nearest confluence cluster
E65.05 INSIDE-confluence flag

E65.06 Score percentile vs the day
E65.07 Cluster freshness (age of youngest member)
E65.08 Score slope as price approaches
E65.09 Confluence co-location with heatmap wall (6h)
E65.10 Top-confluence-of-day flag

O. The Six-Hour Stores (depth.db + trade tape)

Whole-book and per-trade context -- lossless but hard-pruned at 6 hours; recent buckets only.

E66 heatmap columns	6h	Reconstructed resting-liquidity grid (re-anchored at every snapshot, bit-faithful).
E66.01 Nearest wall ABOVE: distance + size		E66.06 SPOOF-PULL flag: wall vanished untouched
E66.02 Nearest wall BELOW: distance + size		E66.07 Approach outcome: held / broke / faded
E66.03 Wall asymmetry (above vs below mass)		E66.08 Wall count within +/- N ticks
E66.04 Wall persistence age		E66.09 Book imbalance around price (resting, not taker)
E66.05 Wall-eaten fraction during the bucket		E66.10 New-wall appearance during bucket flag
E67 BBO series	6h	Best bid/ask through the bucket's lifetime.
E67.01 Mean spread / max spread in bucket		E67.06 Quote-fade events count
E67.02 Spread z vs 6h norm		E67.07 Spread regime tag
E67.03 Microprice drift vs close		E67.08 Spread x volume coupling (wide AND heavy = stress)
E67.04 BBO flip rate (quote instability)		E67.09 Locked/crossed anomalies count
E67.05 Spread at bucket close		E67.10 Spread percentile of the bucket
E68 DOM / COB snapshot	6h	The current-order-book ladder as the terminal's DOM panel reads it.
E68.01 Top-N depth per side		E68.06 Near-touch depth (1-3 ticks)
E68.02 Depth imbalance ratio		E68.07 Pull/stack event counts
E68.03 Depth slope by level (front-loaded vs deep)		E68.08 Depth percentile vs 6h
E68.04 Refresh/replenish rate at touch		E68.09 Ladder skew (bid-heavy vs ask-heavy shape)
E68.05 Large resting order present (Y/N + size)		E68.10 DOM coherence with heatmap wall read
E69 executed trade tape	6h	Every aggTrade (ts, price, qty, side): exact intrabucket sequencing.
E69.01 Intrabucket delta path shape: front / mid / back-loaded		E69.06 Bucket VWAP vs close (who paid up)
E69.02 Largest single print size + its position in the bucket		E69.07 First-third vs last-third delta split
E69.03 Burst count (trades per 100ms clusters)		E69.08 Print-size autocorrelation (iceberg smell)
E69.04 Inter-trade gap stats (mean / max idle)		E69.09 Aggressive-cluster timestamps (sequencing joins)
E69.05 Longest same-side print run		E69.10 Tape-order audit for OHLC-ambiguous barrier hits
E70 trade bubbles / aggregates	6h	Tape rendered as size-scaled bubbles; aggregated by total qty, colored by net side on zoom.
E70.01 Net side per cell		E70.06 Spatial same-side streaks (sweep tracks)
E70.02 Total qty per cell		E70.07 Bubble-at-wall co-location count
E70.03 Max bubble size in bucket		E70.08 Aggregate color flips along price
E70.04 Bubble density (prints per cell)		E70.09 Min-size-filtered bubble count
E70.05 Large-bubble count over threshold		E70.10 Bubble asymmetry above vs below POC
E71 iceberg side classification	6h	Cells flagged iceberg-buy / iceberg-sell via absorption-mark bands.
E71.01 Iceberg cell count + side split		E71.06 First-iceberg-in-window flag
E71.02 Iceberg at bucket extreme flag		E71.07 Iceberg-wall co-location flag
E71.03 Iceberg persistence (recurring at same price)		E71.08 Post-iceberg drift (study join)
E71.04 Iceberg size proxy (qty at flagged cells)		E71.09 Iceberg density regime
E71.05 Coherence with suppression score s		E71.10 Side-switch of icebergs across the bucket
E72 per-cell max buy/sell qty	6h	Largest single executed order per heatmap cell per side.
E72.01 Max buy print / max sell print in bucket		E72.06 Top-cell clustering (adjacent large prints)
E72.02 Max print class vs size_thr anchors (large / whale)		E72.07 Max print share of bucket volume
E72.03 Cells >= large threshold count		E72.08 NEW-max-of-window flag
E72.04 Location of max print within the range		E72.09 Side flip of the max print vs prior bucket
E72.05 Max-print side		E72.10 Max print at wall / at OB co-location

PART II -- Group Derivations: what combinations of entries yield

A group is any 2+ entries read together. The simplest is the canonical example -- `start_time + end_time = elapsed time` -- and the ladder runs up to full multi-family composites. This is where most candidate features for the prediction study live.

G01 -- Elapsed time & clock straddles

members: start_time + end_time

G01.1 Duration (the canonical example: elapsed seconds)

G01.4 Bucket's session-transition position (pre/post-open bucket index)

G01.2 Session-straddle flag (opened and closed in different sessions)

G01.5 Idle gap to the next bucket (tape silence after close)

G01.3 Funding-timestamp straddle flag (crossed 00/08/16 UTC)

G02 -- Candle geometry set

members: open + high + low + close

G02.1 Range, body, wicks, close-position -- the full geometry family

G02.4 Marubozu / doji / pin-bar shape tags

G02.2 Gap vs prior close + true range

G02.5 Equal-highs / equal-lows liquidity-pool distances

G02.3 Inside / engulfing / breakout relation to prior bucket

G02.6 Two-bucket sweep-and-reclaim pattern flags

G03 -- Net initiative

members: buy_vol + sell_vol

G03.1 delta and delta fraction (the canonical pair)

G03.4 Trailing bucket-CVD (cumulative delta)

G03.2 Dominance ratio and log-odds of buy vs sell

G03.5 Buy/sell share slopes (pressure drift)

G03.3 Side-flip flag vs prior bucket

G04 -- Thrust efficiency

members: delta + range (+ body)

G04.1 Price moved per net contract (efficiency of aggression)

G04.4 Efficiency percentile vs window

G04.2 STALL index: heavy delta, no body (absorption smell)

G04.5 Efficiency regime tag

G04.3 OVERDRIVE flag: small delta, big body (vacuum move)

G05 -- Initiative quadrants

members: delta + delta_oI

G05.1 Quadrant: +delta +dOI = new-money push (fresh longs driving)

G05.4 Quadrant: -delta -dOI = unwind drop (longs leaving)

G05.2 Quadrant: +delta -dOI = covering push (shorts forced out)

G05.5 Quadrant streaks and transitions (positioning narrative)

G05.3 Quadrant: -delta +dOI = distribution push (fresh shorts driving)

G06 -- Cohort accounting

members: opL + opS + cIL + cIS + churn

G06.1 Net OI change + open:close ratio

G06.4 Trapped-cohort fuel: opens into a failed break direction

G06.2 Long-cohort net (opL-cIL) and short-cohort net (opS-cIS)

G06.5 Rotation completeness: churn share of total transfer

G06.3 Side initiative index (who is opening harder)

G06.6 OI-story entropy (mixed vs clean narrative)

G07 -- Effort vs result verdict

members: buyer_er + seller_er + body (+ range)

G07.1 ABSORPTION verdict: hot effort, no result

G07.4 Effort-result efficiency ratio per side

G07.2 TRANSLATION verdict: hot effort, full result

G07.5 Verdict streaks (persistent absorption at a level)

G07.3 One-sided-effort divergence (hot side lost the bucket)

G08 -- Urgency

members: *curr_vol + duration (+ avg_velocity)*

G08.1 vel = volume/second and Speed multiple (canonical)

G08.2 Urgency class: heavy AND fast vs heavy-slow grind

G08.3 Pace-adjusted volume anomaly (residual vs both norms)

G08.4 Urgency streak across buckets

G08.5 Urgency at extremes (fast into highs/lows)

G09 -- Squeeze fingerprint

members: *liq_short/liq_long + delta + body*

G09.1 Forced-flow share of the move's aggression

G09.2 Squeeze corroboration: liq side aligned with direction

G09.3 Cascade quality: liq rising across consecutive buckets

G09.4 Post-liq continuation vs fade (study join)

G09.5 Coil read: liqs absorbed with no body (LIQ_COIL)

G10 -- Profile placement

members: *levels + OHLC*

G10.1 poc_pos and close-vs-VA position (acceptance/rejection)

G10.2 Top-vs-bottom loading vs close direction

G10.3 Unfinished-auction extreme vs next-bucket revisit

G10.4 LVN-cross flag: body traversed a low-volume node

G10.5 Profile-shape x direction interaction tags

G11 -- Large-trader read

members: *sz histograms + delta*

G11.1 Large-net vs small-net side split of the bucket's delta

G11.2 DIVERGENCE flag: large one way, small the other

G11.3 Whale-with-move vs whale-against-move tags

G11.4 Large share of the winning side's volume

G11.5 Threshold-sweep robustness of the large read (p90->p97)

G12 -- Large-order placement

members: *sz histograms + levels (+ tape 6h)*

G12.1 Large-print price placement: at extremes vs mid-range

G12.2 Large-at-POC concentration

G12.3 Large prints into the wick (rejection of size)

G12.4 Exact large-print sequencing within the bucket (6h tape)

G12.5 Large-print co-location with imbalance stacks

G13 -- Who absorbed whom

members: *suppression s + delta sign (+ levels)*

G13.1 Absorbed-side attribution: aggressors who failed

G13.2 Absorbing-level location (where the wall stood)

G13.3 Validated absorption against the trend flag

G13.4 Absorption-then-flip candidate sequence (study join)

G13.5 s x stacked-imbalance coherence check

G14 -- Toxic push composite

members: *vpin + delta fraction + vol_mult*

G14.1 Toxic AND one-sided AND fast composite score

G14.2 Informed-flow candidate flag (all three elevated)

G14.3 Composite percentile vs window

G14.4 Composite streak (sustained informed pressure)

G14.5 Composite at structure (at OB / at wall) join

G15 -- State sequencing

members: *state verdict + prior/next buckets*

G15.1 Transition-pair features (e.g. COIL -> SQUEEZE)

G15.2 State run-length at current bucket

G15.3 Verdict-flip density (regime instability)

G15.4 Confidence-weighted transition score

G15.5 Per-transition outcome stats vs the barrier labels (study join)

G16 -- Window normalization family

members: any metric + trailing 10/30/240 windows

G16.1 z-score version of every raw metric

G16.2 Percentile version of every raw metric

G16.3 Streak length above/below own baseline

G16.4 Slope of every metric over last N buckets

G16.5 Regime tag per metric (calm/normal/hot)

G17 -- Structure-conditioned bundle

members: bucket + order_blocks (+ confluence)

G17.1 At-OB behavior: touch / reject / break with all flow features

G17.2 Distance-decay features (flow x proximity to OB)

G17.3 Confluence-weighted position score

G17.4 First-touch vs n-th-touch interaction with absorption

G17.5 OB-break with large-order participation flag

G18 -- Wall interaction bundle

members: bucket + heatmap walls (6h) -- the untested angle

G18.1 Approach-outcome label: wall held / broke / pulled

G18.2 Wall-eaten rate vs taker aggression (who is winning)

G18.3 Absorption s AT a visible wall (the flagged study)

G18.4 Spoof-pull preceding the move flag

G18.5 Wall distance as a barrier-odds conditioning feature

G19 -- Sequencing bundle

members: bucket + trade tape (6h)

G19.1 Front-loaded vs back-loaded aggression shape

G19.2 Largest-print timing vs bucket outcome

G19.3 Burst-timing pattern (early bursts vs late bursts)

G19.4 Same-side run lengths inside the bucket

G19.5 Exact barrier-hit ORDER for OHLC-ambiguous buckets (labeler upgrade)

G20 -- Cross-timeframe join

members: 1m bucket + live 5m/15m/1h/4h context

G20.1 Position of the 1m bucket inside the forming higher-tf bucket

G20.2 Higher-tf state verdict at the moment

G20.3 Higher-tf delta/dOI alignment flag

G20.4 Multi-tf agreement score (how many tfs point the same way)

G20.5 Higher-tf OB/confluence proximity at the 1m close

G21 -- Trap / squeeze composite

members: exhaustion + sweep + reclaim + trapped-opens

G21.1 The state-engine trap recipe exposed as one composite feature

G21.2 Sweep-fail-reclaim three-beat pattern flag

G21.3 Trapped-cohort size behind the failed break

G21.4 Squeeze-completion score (reclaim + forced flow + speed)

G21.5 Composite vs the labeled outcome (study join)

G22 -- Effort triangle

members: duration + range + curr_vol

G22.1 Volatility per unit volume (fragility of the tape)

G22.2 Time-price-volume consistency residual (anomaly detector)

G22.3 Slow-wide flag (thin tape, big moves)

G22.4 Fast-narrow flag (heavy grind, no travel)

G22.5 Triangle regime tag joining pace and volatility

PART III -- The Mode-10 Selection Feature Set: 11 panels + selection stats, under THREE scopes

For every studied bucket the study draws a Mode-10 selection over **the last 15 buckets plus the entry bucket** and extracts the exact value of every panel and the selection stats box. The SAME field set is then re-extracted on two post-outcome selections. Codes multiply by scope prefix: **B-P4.02** is the base-selection value, **J-P4.02** the journey value, **X-P4.02** the excursion value.

B- **BASE selection** -- The 15 buckets BEFORE entry + the entry bucket (16 total), extracted at entry close.

ENTRY-LEGAL

J-	JOURNEY selection -- Entry bucket through the bucket that TOUCHED the barrier (TP or SL).	POST-HOC
X-	EXCURSION selection -- Entry bucket through the bucket of the DIRECTIONAL 6h max: TP episodes end at the max IN the position's favor (a long that ran to +1.0% ends there); SL episodes end at the max adverse point.	POST-HOC

P1 -- ABSORPTION panel

Bull vs bear absorbed volume (V^ s) across the selection; badges carry the side percentages.*

P1.01 Bull absorption percentage (badge)	P1.07 Sum: absorbed share of selection volume
P1.02 Bear absorption percentage (badge)	P1.08 Trajectory slope across the selection
P1.03 Spread: bull% - bear%	P1.09 Peak-absorption bucket position (0..15)
P1.04 Dominant side label	P1.10 Validated-dot ($s \geq 0.60$) count in selection
P1.05 Absolute bull V^* s over the selection	P1.11 Entry-bucket raw (un-smoothed) hover value
P1.06 Absolute bear V^* s over the selection	

P2 -- EFF-AGG panel

Bull vs bear effective aggression $V^(1-s)$ -- the mirror partition.*

P2.01 Bull eff-agg percentage (badge)	P2.07 Sum: eff-agg share of selection volume
P2.02 Bear eff-agg percentage (badge)	P2.08 Trajectory slope across the selection
P2.03 Spread: bull% - bear%	P2.09 Peak eff-agg bucket position
P2.04 Dominant side label	P2.10 Dominance-flip count across the selection
P2.05 Absolute bull $V^*(1-s)$ over the selection	P2.11 Entry-bucket raw value
P2.06 Absolute bear $V^*(1-s)$ over the selection	

P3 -- E/R panel

Buyer vs seller effort-per-result across the selection.

P3.01 Buyer-side percentage (badge)	P3.06 Aggregate seller multiple vs baseline
P3.02 Seller-side percentage (badge)	P3.07 Hot-bucket count ($\geq 1.5x$) in selection
P3.03 Spread: buyer% - seller%	P3.08 Both-hot bucket count
P3.04 Dominant side label	P3.09 Peak-E/R bucket position
P3.05 Aggregate buyer multiple vs trailing baseline	P3.10 Entry-bucket raw value

P4 -- EXHAUSTION panel

Gated exhaustion with side badge -- the value + last-exhausted-side readout.

P4.01 Current exhaustion value (badge)	P4.07 Bear-fire count inside the selection
P4.02 LAST EXHAUSTED SIDE (bull / bear)	P4.08 Buckets since the last fire
P4.03 Last-exhausted side's percentage	P4.09 Peak gated score + its bucket position
P4.04 Opposite side's percentage	P4.10 Crossover-diamond count in the selection
P4.05 Spread between the two sides	P4.11 Weakest gating leg at the entry bucket
P4.06 Bull-fire count inside the selection	

P5 -- Phase panel: BEFORE

Two lines = the BEFORE phase's bull/bear trajectory (default-ON phase panel).

P5.01 Phase bull percentage	P5.06 Phase length in buckets
P5.02 Phase bear percentage	P5.07 Phase trajectory slope
P5.03 Spread: bull% - bear%	P5.08 Phase net delta
P5.04 Phase winner (UP green / DOWN red)	P5.09 Phase intensity vs selection average
P5.05 Phase share of selection volume	P5.10 Entry-membership flag (entry inside this phase)

P6 -- Phase panel: DURING

The START/DURING merged phase's bull/bear trajectory.

P6.01 Phase bull percentage	P6.06 Phase length in buckets
P6.02 Phase bear percentage	P6.07 Phase trajectory slope
P6.03 Spread	P6.08 Phase net delta
P6.04 Phase winner	P6.09 DURING-minus-BEFORE shift (phase transition)
P6.05 Phase share of selection volume	P6.10 Phase peak bucket position

P7 -- Phase panel: END

The END phase's bull/bear trajectory -- the freshest read at entry.

P7.01 Phase bull percentage	P7.06 Phase length in buckets
P7.02 Phase bear percentage	P7.07 Phase trajectory slope
P7.03 Spread	P7.08 Phase net delta
P7.04 Phase winner	P7.09 END-minus-DURING shift (late acceleration)
P7.05 Phase share of selection volume	P7.10 END-winner alignment with entry-bucket direction

P8L -- LARGE MKT panel

Market-order size class LARGE ('8' cycle slot 1) -- the broad daemon threshold, not selection-local.

P8L.01 Large BUY volume in selection	P8L.07 Largest print class vs size_thr anchors
P8L.02 Large SELL volume in selection	P8L.08 Large-activity slope across selection
P8L.03 Sum: total large volume	P8L.09 Run-winner badge total (latest bar)
P8L.04 Large delta (buy - sell)	P8L.10 Threshold in force (slider or p95 auto)
P8L.05 Large share of selection volume	P8L.11 Large bull/bear split at the entry bucket
P8L.06 Large trade count	

P8S -- SMALL MKT panel

Market-order size class SMALL ('8' cycle slot 2).

P8S.01 Small BUY volume in selection	P8S.06 Small trade count
P8S.02 Small SELL volume in selection	P8S.07 Small-activity slope across selection
P8S.03 Sum: total small volume	P8S.08 Small-crowd dominant side
P8S.04 Small delta (buy - sell)	P8S.09 LARGE-vs-SMALL divergence flag (opposed sides)
P8S.05 Small share of selection volume	P8S.10 Small threshold in force (~p50)

P9 -- COMPOSITE LEAN panel

One line per side = per-bucket average of the four measure panels' signed leans (+/- own-side exhaustion); badges = BULL / BEAR / SUM.

P9.01 Bull line value (badge)	P9.06 Sum slope over the last 5 buckets
P9.02 Bear line value (badge)	P9.07 Zero-cross count in the selection
P9.03 SUM line value (badge)	P9.08 +/-50 breach count
P9.04 Spread: bull - bear	P9.09 Mean lean across selection (conviction)
P9.05 Sum sign (net lean direction)	P9.10 Entry-bucket raw lean

P0 -- SMOOTHED LEAN panel

Panel 9's smoothed twin: (current + locked)/2, locked region, settling tail, and confirmed +50/0/-50 cross markers.

P0.01 Smoothed bull value	P0.07 Last confirmed cross level (+50 / 0 / -50)
P0.02 Smoothed bear value	P0.08 Side of the last confirmed cross
P0.03 Smoothed SUM value	P0.09 Buckets since the last confirmed cross
P0.04 Spread	P0.10 Cross-marker count in the selection
P0.05 Locked-region end value	P0.11 Twin-coherence flag (sign agrees with P9)
P0.06 Settling-tail delta (current - locked)	

S -- Selection stats box (the Mode-10 readout, aggregated over the selection)

S.01 Selection bucket count	S.08 Sum OpS
S.02 Selection wall-clock duration	S.09 Sum CIL
S.03 Total volume	S.10 Sum CIS
S.04 Sell Buy split	S.11 Aggregate B-E/R multiple
S.05 Net delta + delta percentage	S.12 Aggregate S-E/R multiple
S.06 Net dOI	S.13 Average VEL + Speed multiple
S.07 Sum OpL	S.14 Selection range + net price change %

C -- Last-15 pre-entry context (single scope, ENTRY-LEGAL)

The window's own story before the entry: was price going up or down, and how.

C.01 Direction over the 15: net UP or DOWN	C.07 Higher-highs / lower-lows counts
C.02 Net % change over the 15	C.08 Dominant 12-state verdict over the 15
C.03 Linear-fit slope + R-squared (trendiness)	C.09 Cumulative delta over the 15
C.04 Path class: trend / range / V-shape	C.10 Cumulative dOI over the 15
C.05 Max drawup and drawdown within the 15	C.11 Compression share (buckets under 0.6 ATR)
C.06 Realized volatility of the 15	C.12 Sweep-event count in the 15

PART IV -- Outcome & Excursion (the label side) + external baselines

O.* fields ARE the outcomes and their post-hoc descriptors. **Directional rule:** TP episodes measure the excursion only IN the position's favor (a long that hit TP and ran to +1.0% within the 6h window is studied to that +1.0% max and no further); SL episodes measure only the adverse continuation. All excursions cap at the 6h window; a max sitting exactly at the edge is flagged censored.

L -- Episode identity & labels (the row's spine; one row per bucket x direction)

L.01 Episode direction: long / short	L.06 Entry bucket id + entry timestamp
L.02 Outcome label: TP / SL / UNRESOLVED	L.07 Ambiguous-hit flag: one bucket spanned BOTH barriers -> ruled SL by convention
L.03 Resolved flag: barrier hit within the 6h window and available data	L.08 Joint 3-way path label mirror (UP / DOWN / WHIPSAW)
L.04 Censor reason: none / 6h-window / end-of-data	L.09 Episode id: bucket_id x direction key
L.05 Entry price (entry-bucket close) + absolute TP and SL prices	

O -- Barrier outcome + excursion family (POST-HOC: labels and descriptors, never entry features)

O.01 Three-way path label: UP-resolve / DOWN-resolve / WHIPSAW	O.14 Adverse velocity: % per hour
O.02 Per-direction outcome: long TP/SL and short TP/SL	O.15 Recovery flags: re-crossed SL / re-crossed entry within 6h
O.03 Time to barrier touch (entry close -> touch)	O.16 Excursion-max bucket id (end of X-scope)
O.04 Buckets to barrier touch	O.17 TP episodes: max against-entry depth BEFORE the hit, % from entry
O.05 Barrier-touch bucket id (end of J-scope)	O.18 SL episodes: max in-favor move BEFORE the stop, % from entry
O.06 TP episodes: max favorable excursion BEYOND TP, % (6h cap)	O.19 TP episodes: price crossed entry AGAINST the position first (Y/N)
O.07 TP episodes: time to that max	O.20 TP episodes: underwater time before the TP hit
O.08 TP episodes: buckets to that max	O.21 TP episodes: near-death ratio -- against-entry depth / SL distance
O.09 Excursion velocity: % per hour to the max	O.22 SL episodes: price crossed entry IN FAVOR first (Y/N)
O.10 Retention: window-end value vs max (give-back %)	O.23 SL episodes: time in profit before the stop
O.11 Censoring flag: max sits at the 6h window edge	O.24 SL episodes: near-win ratio -- in-favor move / TP distance
O.12 SL episodes: max adverse continuation BEYOND SL, %	O.25 Entry re-cross count before the barrier (chop around entry)
O.13 SL episodes: time to the max adverse point	

K -- Keltner channel & POC baseline positioning (ENTRY-LEGAL, computed offline)

NOT terminal overlays -- the study computes both from bucket OHLC/levels. Defaults frozen: $KC = EMA-20 \pm 2.0 \times ATR-20$; POC baseline = 240-bucket rolling POC. If your chart uses different settings, give Claude Code the numbers BEFORE the run.

K.01 KC middle line value at entry (EMA-20 of close)	K.07 Distance to nearest band (ticks and ATRs)
K.02 KC upper band (mid + $2.0 \times ATR-20$)	K.08 Channel slope (mid-line drift over N buckets)
K.03 KC lower band (mid - $2.0 \times ATR-20$)	K.09 Squeeze flag: bandwidth in bottom percentile
K.04 Bandwidth: (upper - lower) / mid	K.10 Rolling-POC baseline value (240-bucket window POC)
K.05 Entry close position inside the channel, 0..1	K.11 Close vs POC baseline: side + distance
K.06 Zone tag: above / upper-half / lower-half / below	K.12 POC-baseline slope + consecutive-side streak

Availability & Honesty Notes -- read before any study

Leakage guard (critical). J- and X-scope fields and every O.* field contain POST-ENTRY information. They may be compared between the TP and SL groups to understand HOW outcomes unfold -- but they must NEVER enter an entry-decision rule or model. The entry-legal universe is: E*, G*, C.*, K.*, and B- scope only. The scope prefix makes this mechanically enforceable in the study code.

History depth. closed_buckets caps at 10,000 buckets per timeframe (~15 days at the 1m pace). Raising the cap is queued now that disk cost is trivial.

Field ages. liq_* is structurally zero before the liquidation feed; sz_* exists only from 2026-06-30 (so B-P8L / B-P8S coverage starts there). Studies must MASK by availability -- zeros there are absence of data.

The 6-hour wall. Everything tagged (6h) prunes at 6 hours. Full-history passes use bucket-persisted fields; wall/tape features need retention extended FORWARD first.

Bit-identical recomputation. Every derived, windowed and panel item is a pure function of persisted fields using the repo's own code (the Mode-10 measures recompute offline exactly).

The multiplicity warning. This registry now codes 1281 fields against only a few hundred independent barrier episodes. The study PRE-REGISTERS a small entry-legal subset, judges it on a time-split holdout touched once, against the 37.5% null and the ~48.8% fee-adjusted breakeven. Mining the whole menu in-sample guarantees false patterns; the registry exists so selection is deliberate.